

Cycle 3 - Two-Sample Inference

Q8: SadOrHopeless Feeling and Current Cigarette Use

Research Question

Is the proportion of current cigarette use different between students who felt sad or hopeless and those who did not?

11.43 pp

higher current cigarette use in the SadOrHopeless group

Key statistical result

Method: two-proportion z-test

95% CI: [9.83, 13.02] pp

z: 15.0536 **p-value:** < 0.0001

Decision: Reject H0 at $\alpha = 0.05$

Conclusion

Students who felt sad or hopeless had a higher observed proportion of current cigarette use (27.6% vs 16.2%). The result is statistically significant, but it shows association, not causation.

